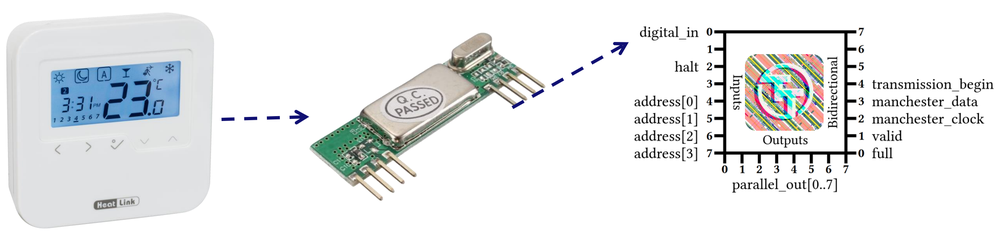
Decoding Manchester Coded Transmissions in Hardware

Zachary Kohnen, Alex Alvarado

PROJECT AIM

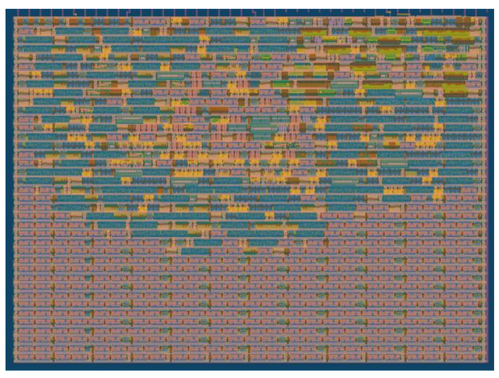
Demonstrate a SkyWater 130nm ASIC based decoder for Manchester coding within one TinyTapeout tile in less than 16 weeks



Thermostat | RXB6 OOK Receiver | TinyTapeout ASIC

IMPLEMENTATION

- Open-source skywater 130nm PDK provided by TinyTapeout. [1]
- Synthesized in one tile with the support from TinyTapeout's online community.
- With all optimizations, the final utilization was reduced from 62% to 32% [2]



2D render of ASIC layers

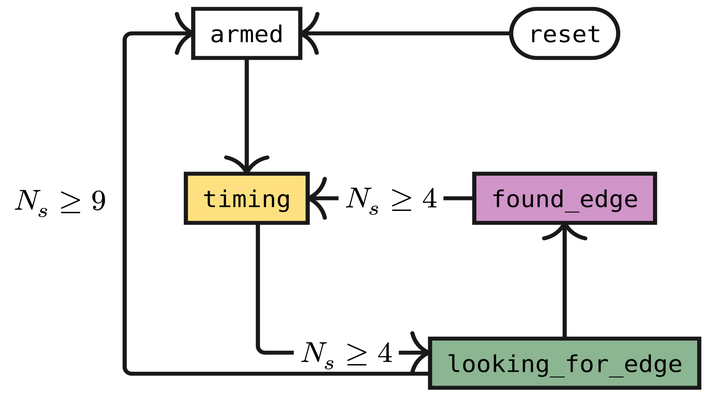


Logic analyzer (left) and TinyTapeout carrier board (right) under test

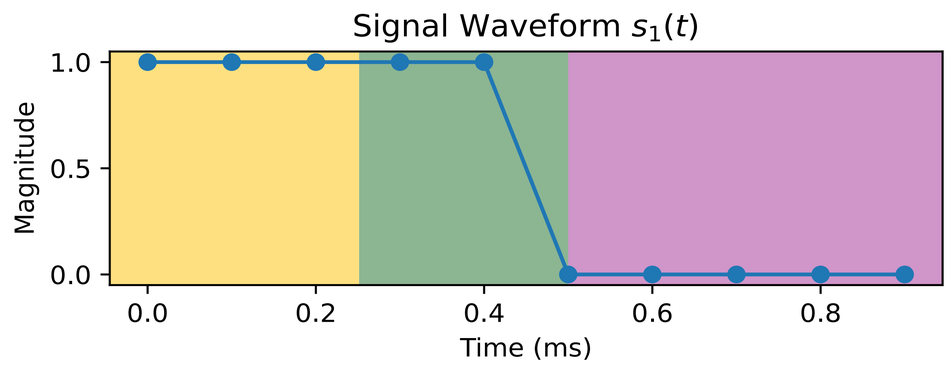
TAKEAWAY

- Testing ASIC designs in simulation with real-world data is important
- ASIC design and tape-outs are accessible, even to those with no prior experience.
- ASIC design requires different considerations than FPGA design but having experienced community members helps accelerate learning.

DESIGN

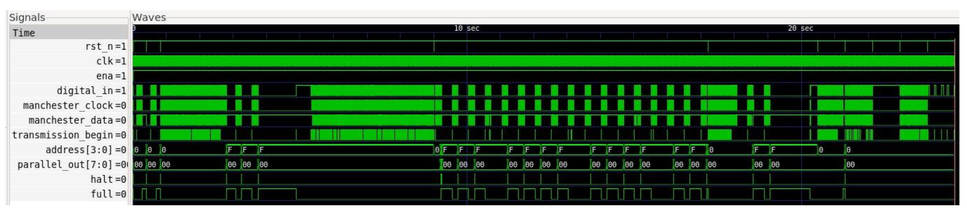


- Finite state machine to detect data edges in Manchester coding, filtering inter-symbol edges
- Implemented using oversampling and a timer-based approach.



TESTING

- Transmissions from the thermostat were captured with the RXB6
- During development these transmissions were replayed into simulations.
- Replayed transmissions were tested with a logic analyzer successfully once the chip was received



REFERENCES

1. M. Venn, "Tiny Tapeout: A shared silicon tape out platform accessible to everyone," in IEEE Solid-State Circuits Magazine, vol. 16, no. 2, pp. 20-29, Spring 2024, doi: 10.1109/MSSC.2024.3381097
2. Z. Kohnen, "Utilizing Tiny Tapeout for Low-Cost ASIC Development," B.S. thesis, ICT Lab, Dept. of Elect. Eng., Eindhoven Univ. of Technology, 2023.

Information and Communication Theory Lab
Signal Processing Systems Group
Future of Chips Event

Conference Presentation

GitHub